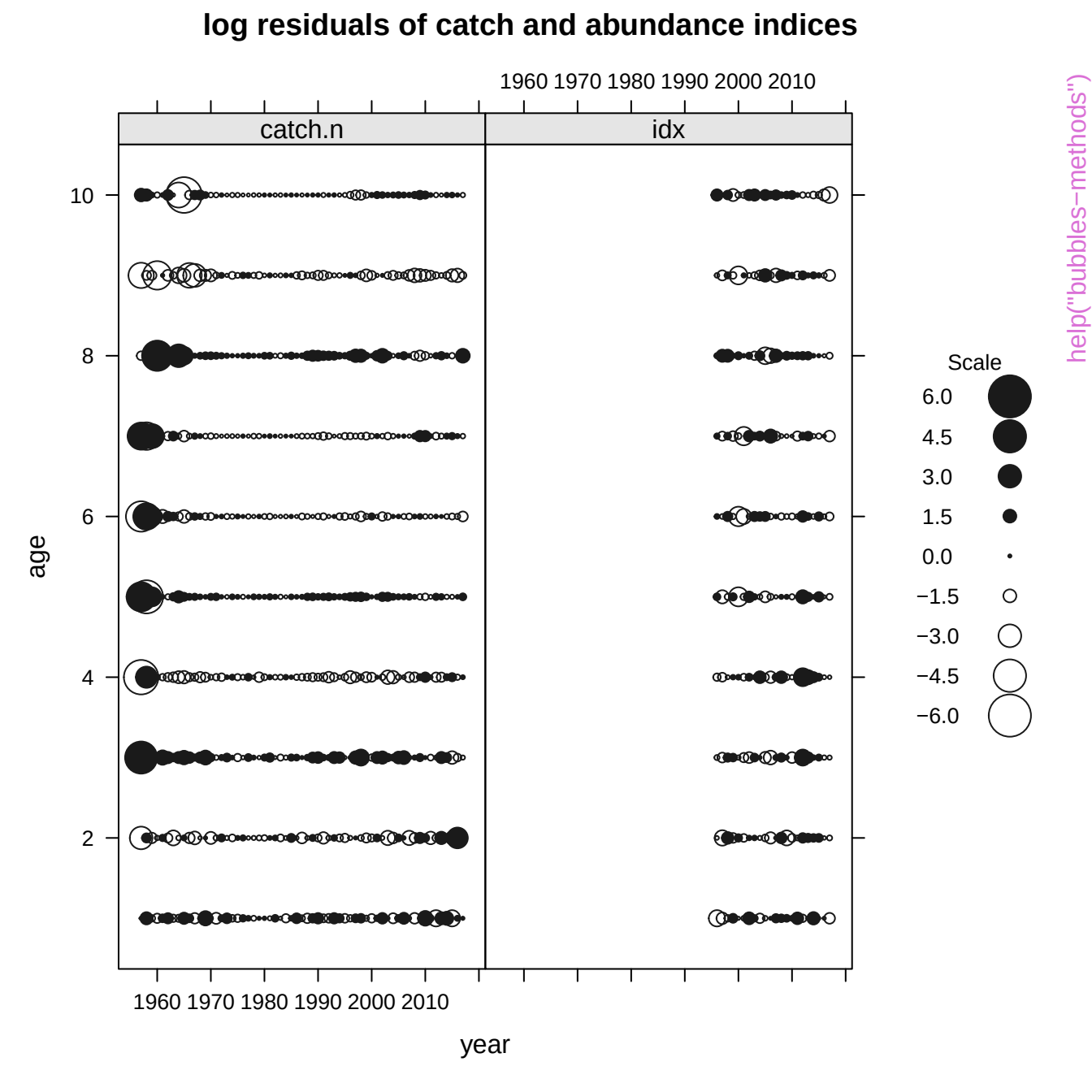
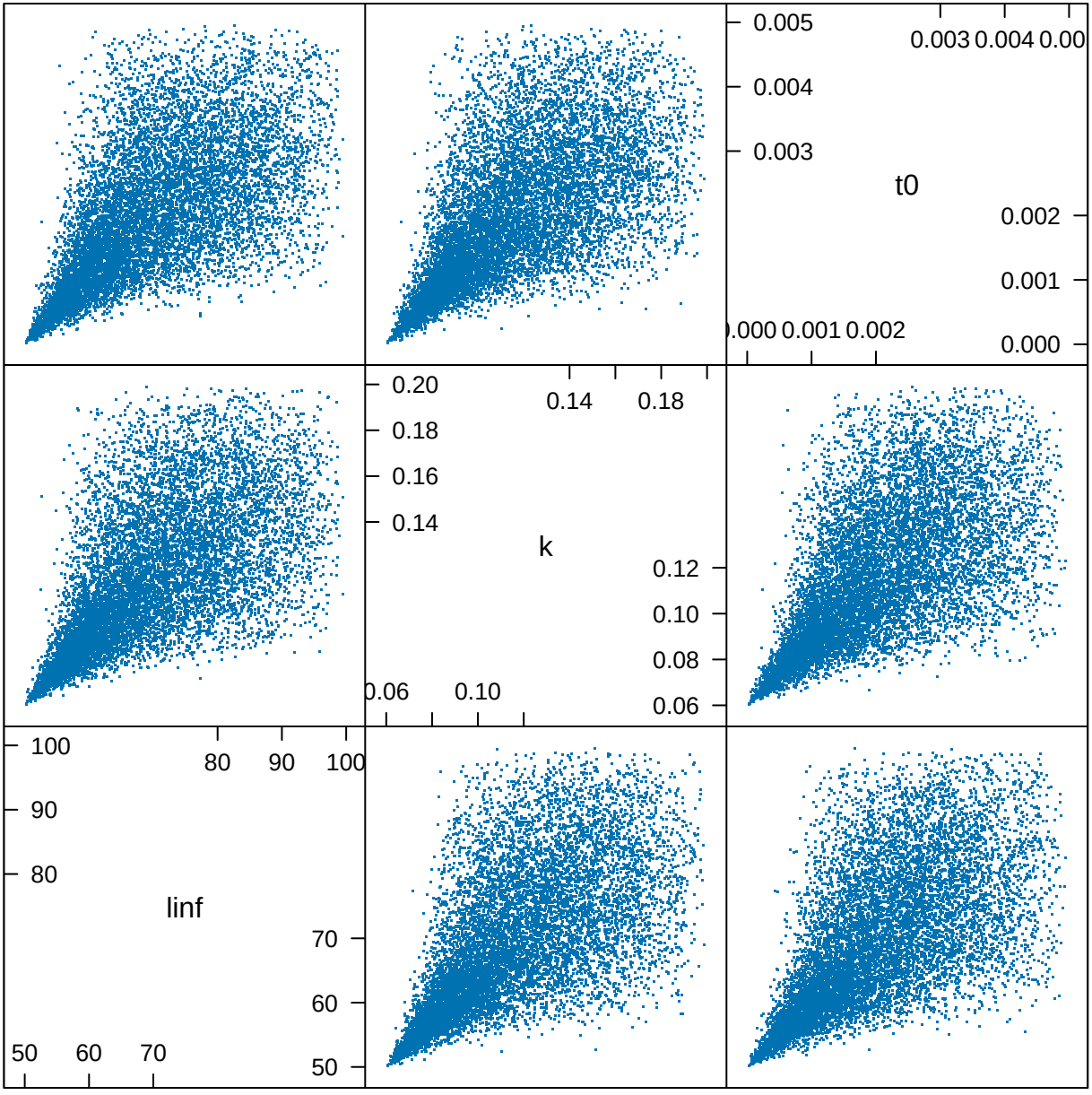
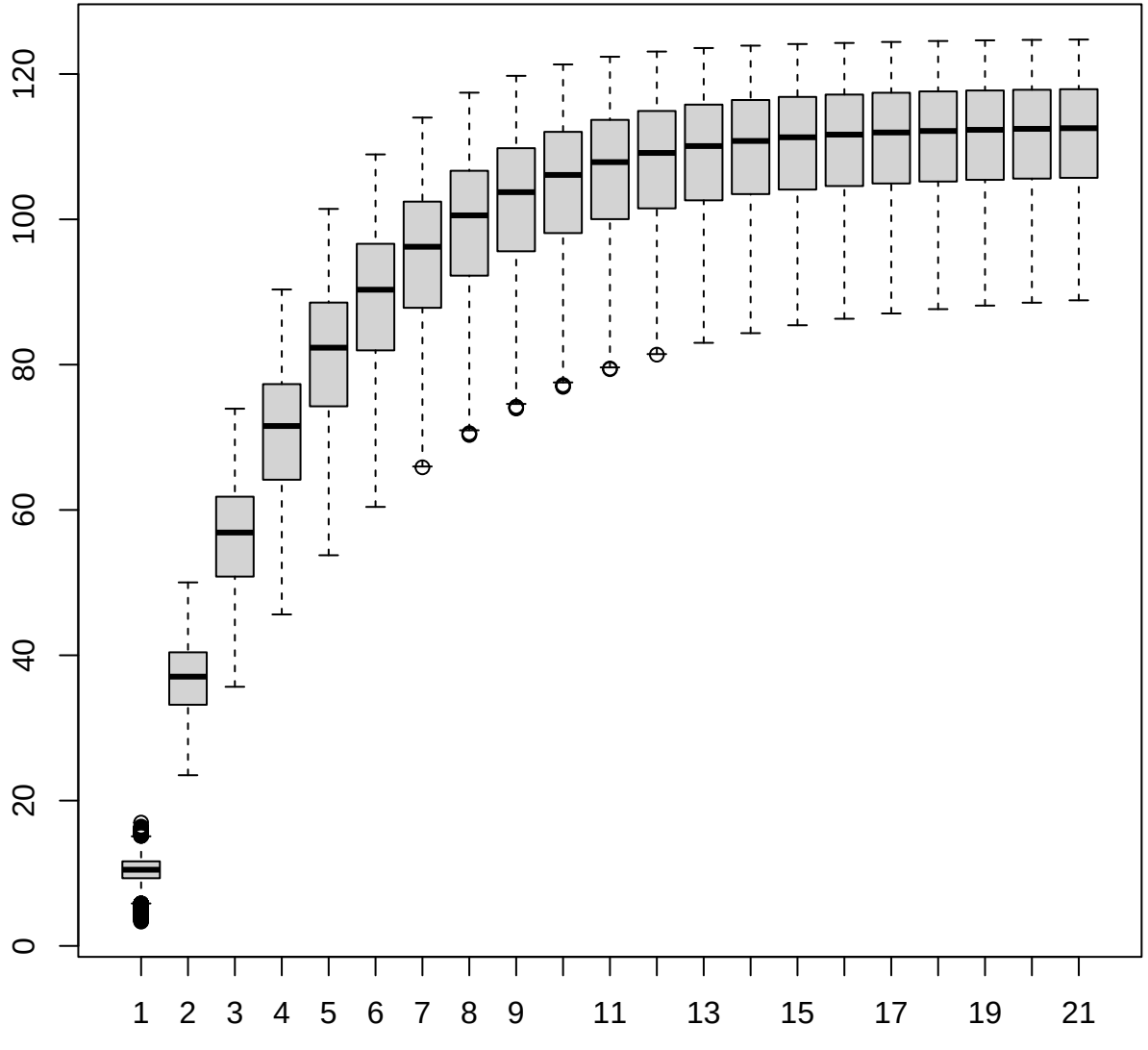

log residuals of catch and abundance indices
1960 1970 1980 1990 2000 2010
catch.n
idx
10
8
6
4
2
age
1960 1970 1980 1990 2000 2010
year
Scale
6.0
4.5
3.0
1.5
0.0
−1.5
−3.0
−4.5
−6.0
help("bubbles−methods")

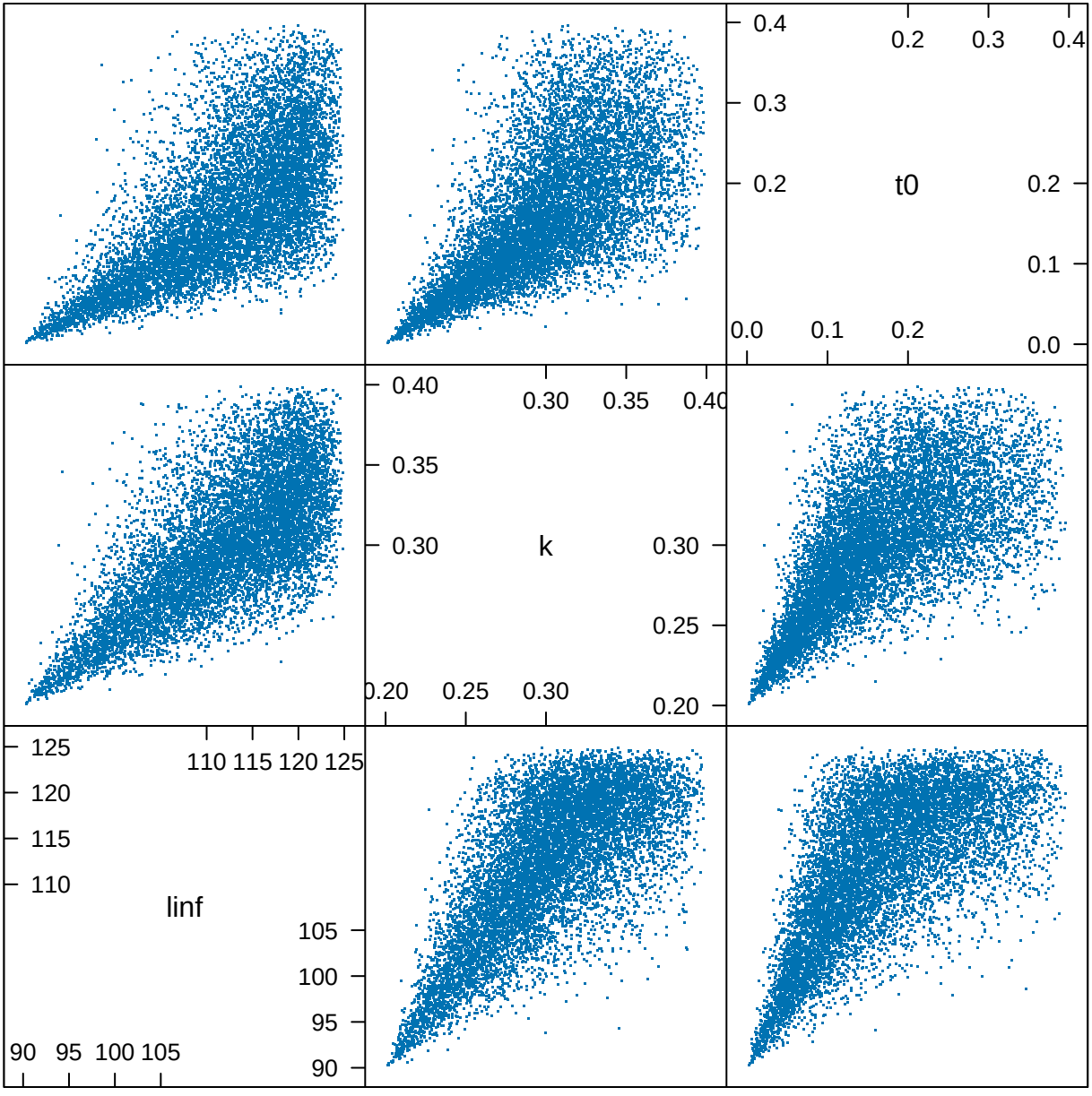


Scatter Plot Matrix

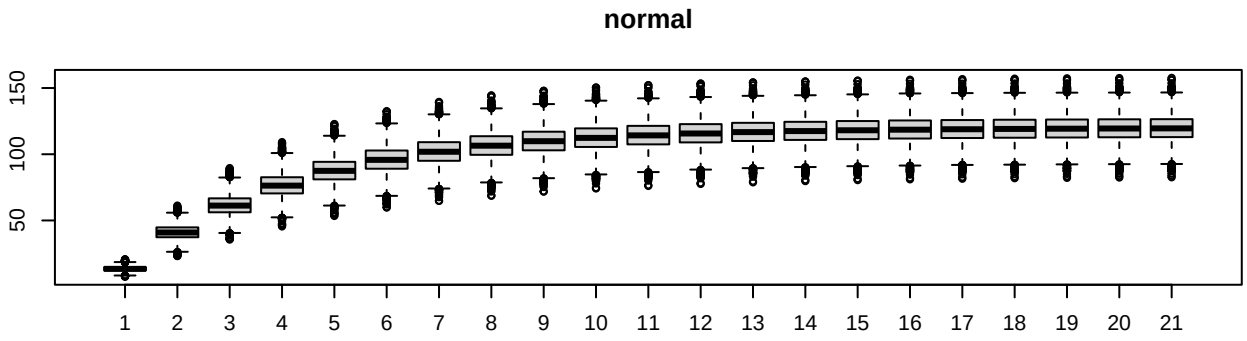
help("mvrcop-a4aGr")



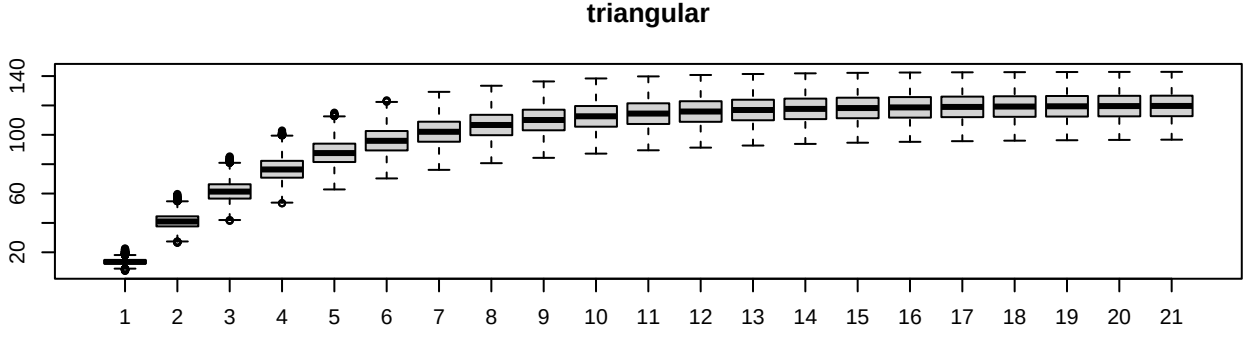
help("mvrcop")

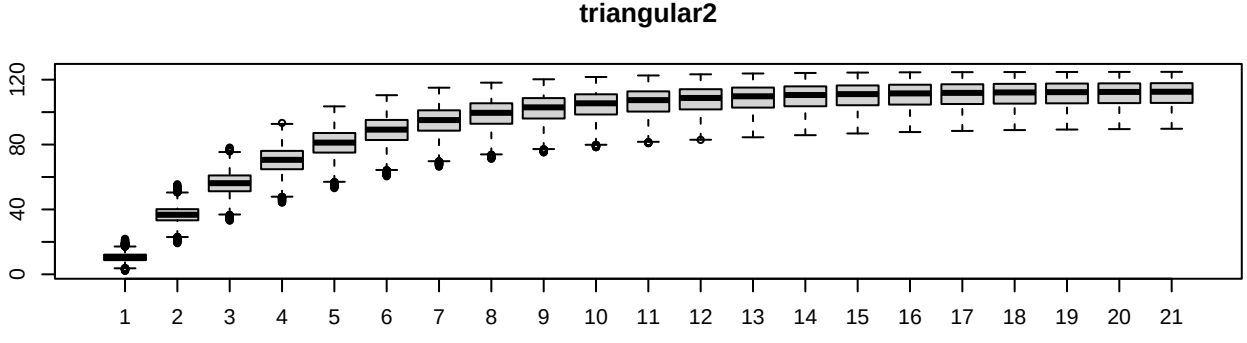


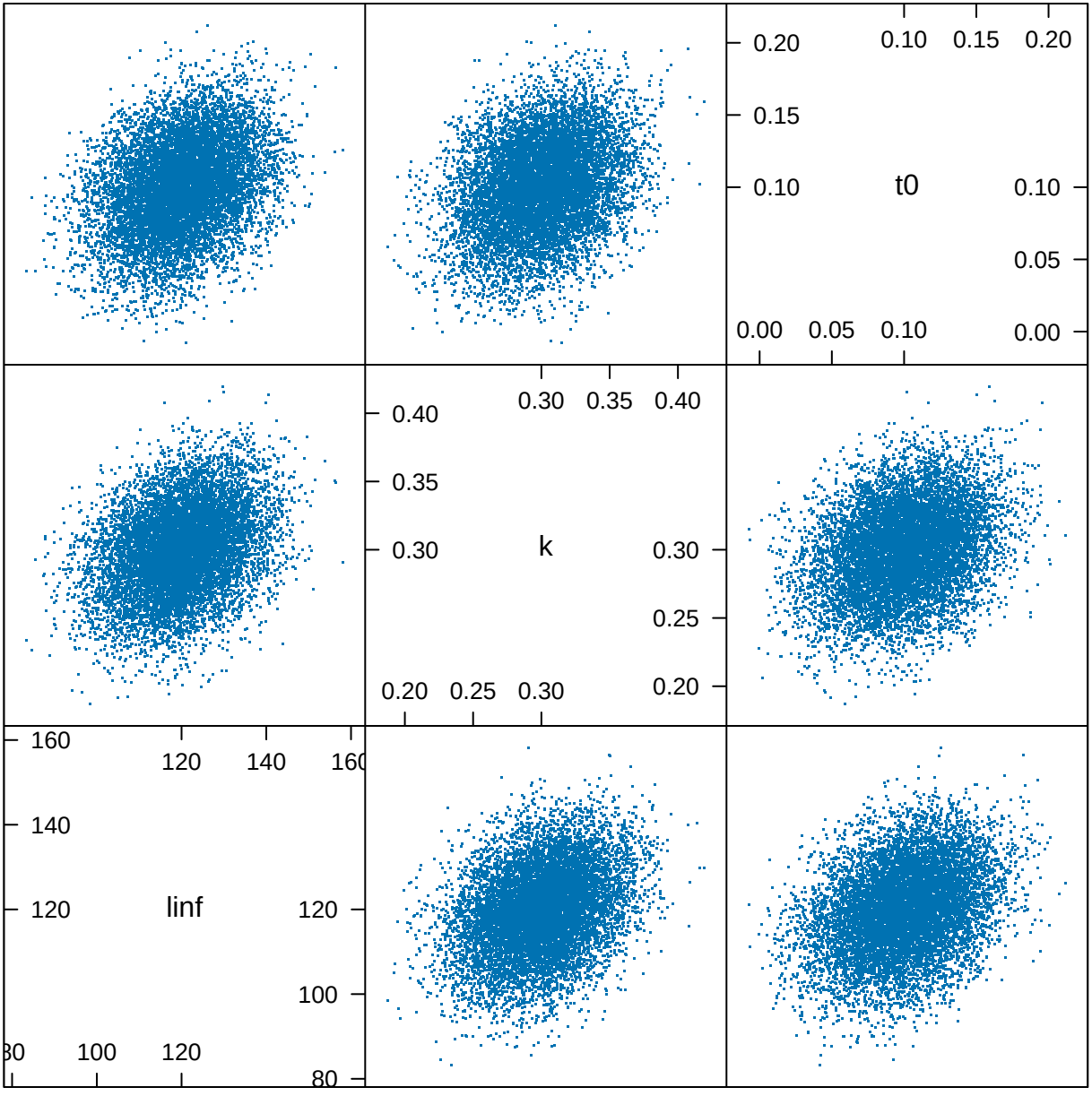
help("mvrcop")



help("mvrtriangle")

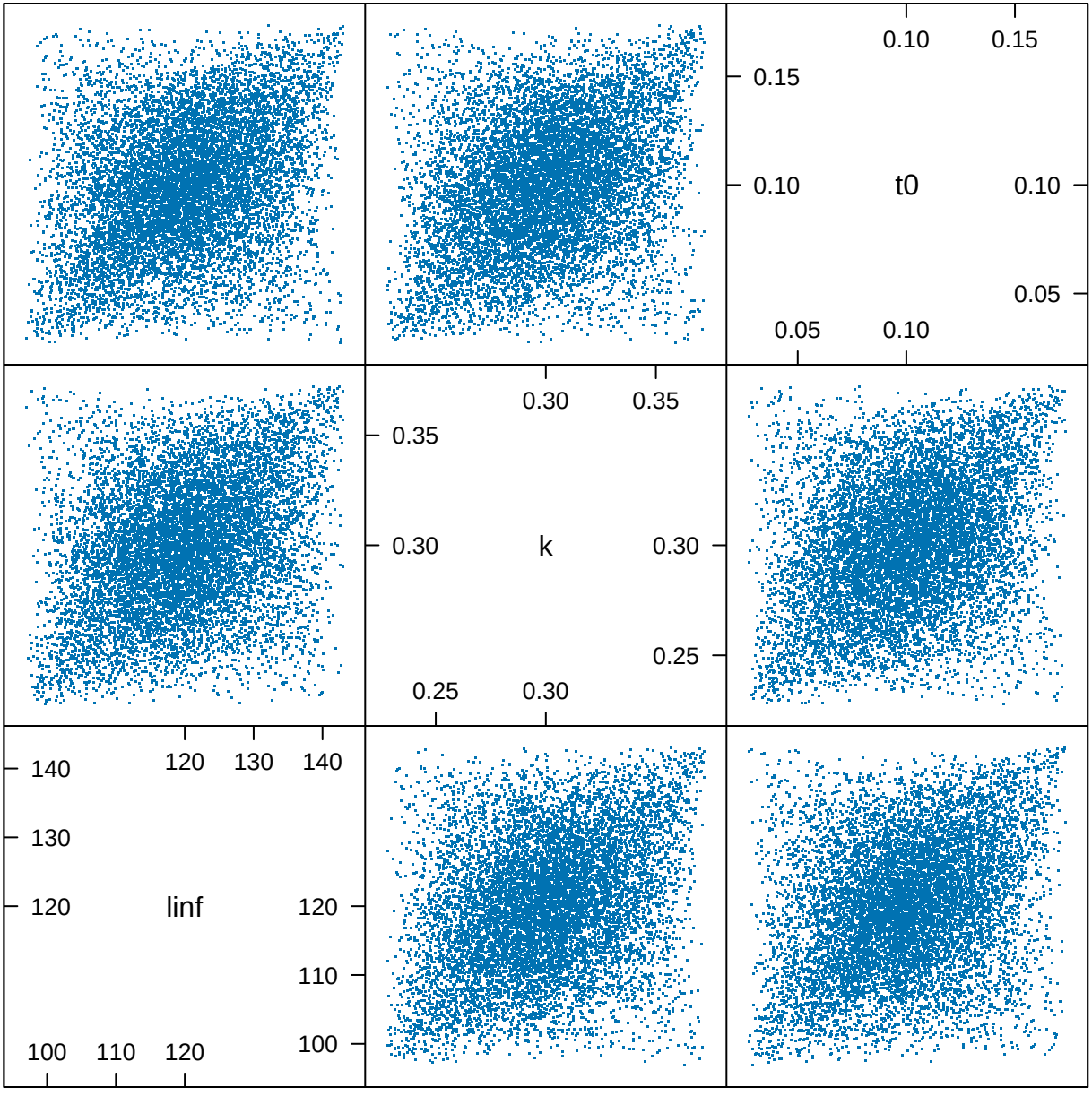






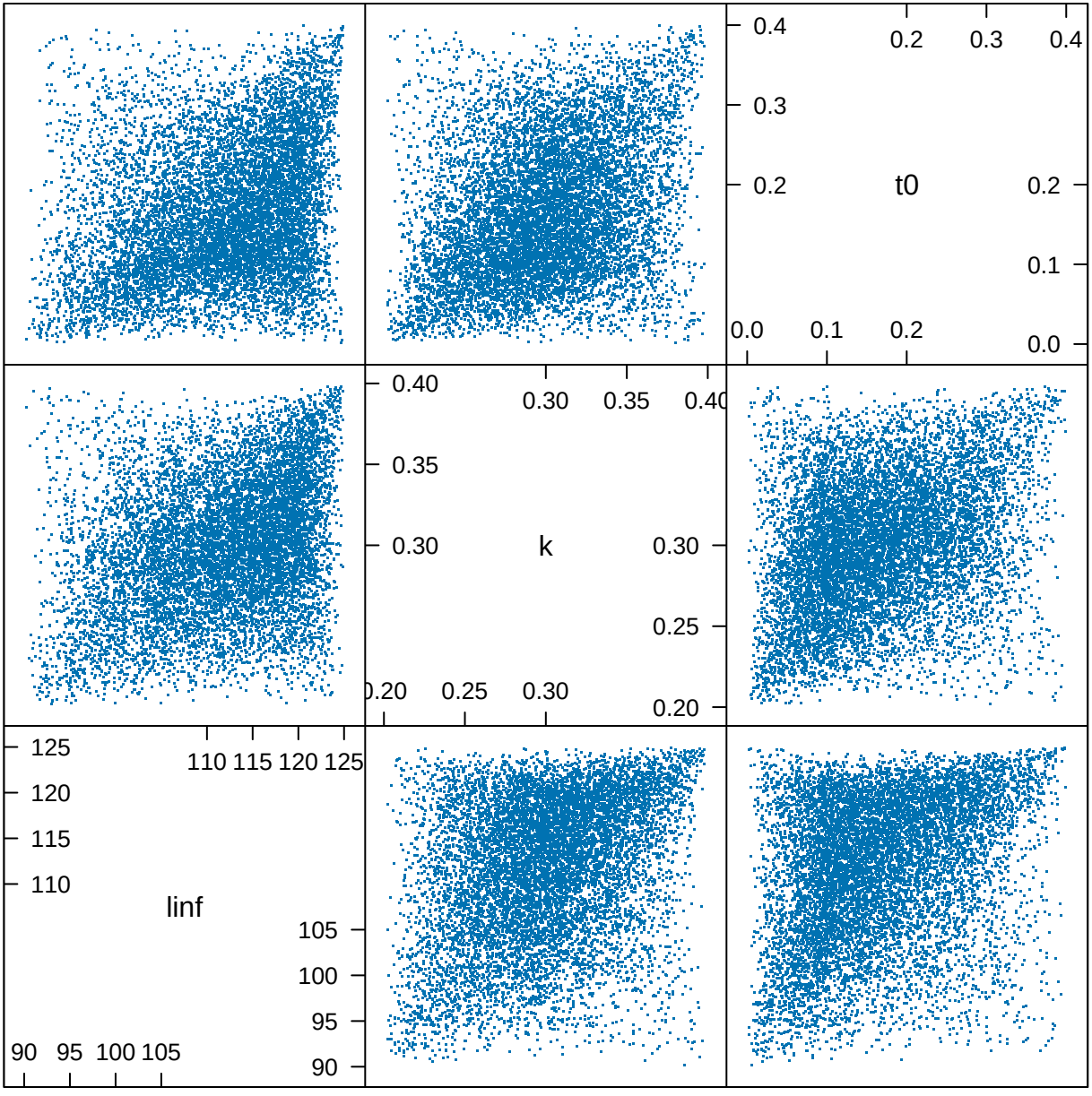
Scatter Plot Matrix

help("mvrtriangle")



Scatter Plot Matrix

help("mvrtriangle")



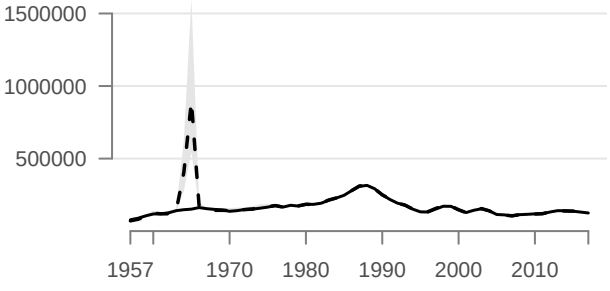
Scatter Plot Matrix

help("mvrtriangle")

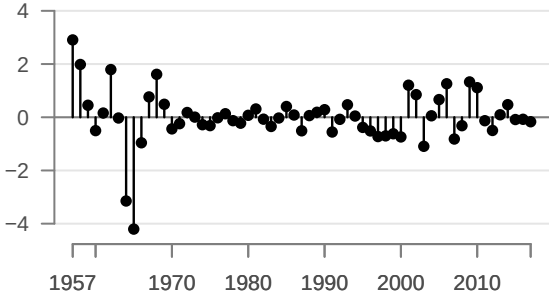
Aggregated catch diagnostics

help("plot-catch")

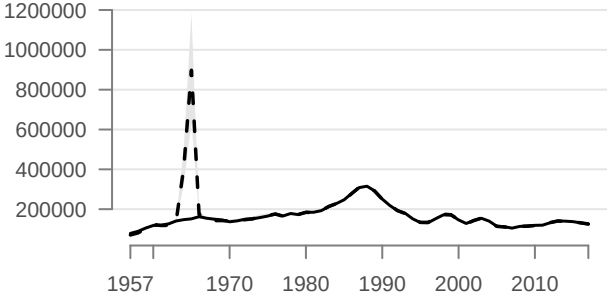
Observation error



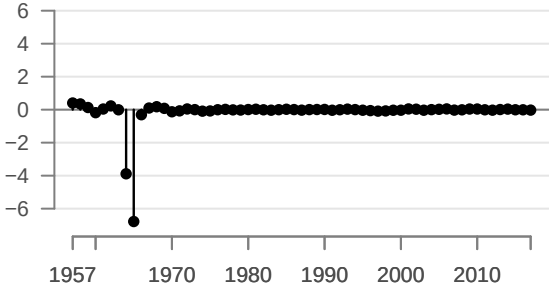
Pearson residuals



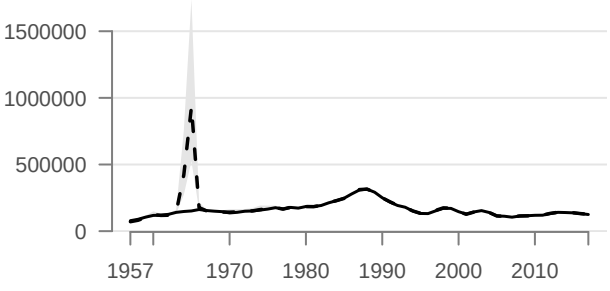
Estimation error



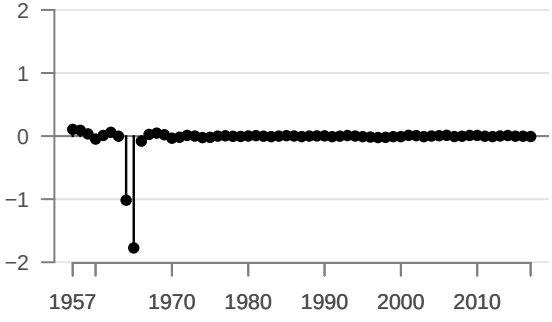
Standardized residuals



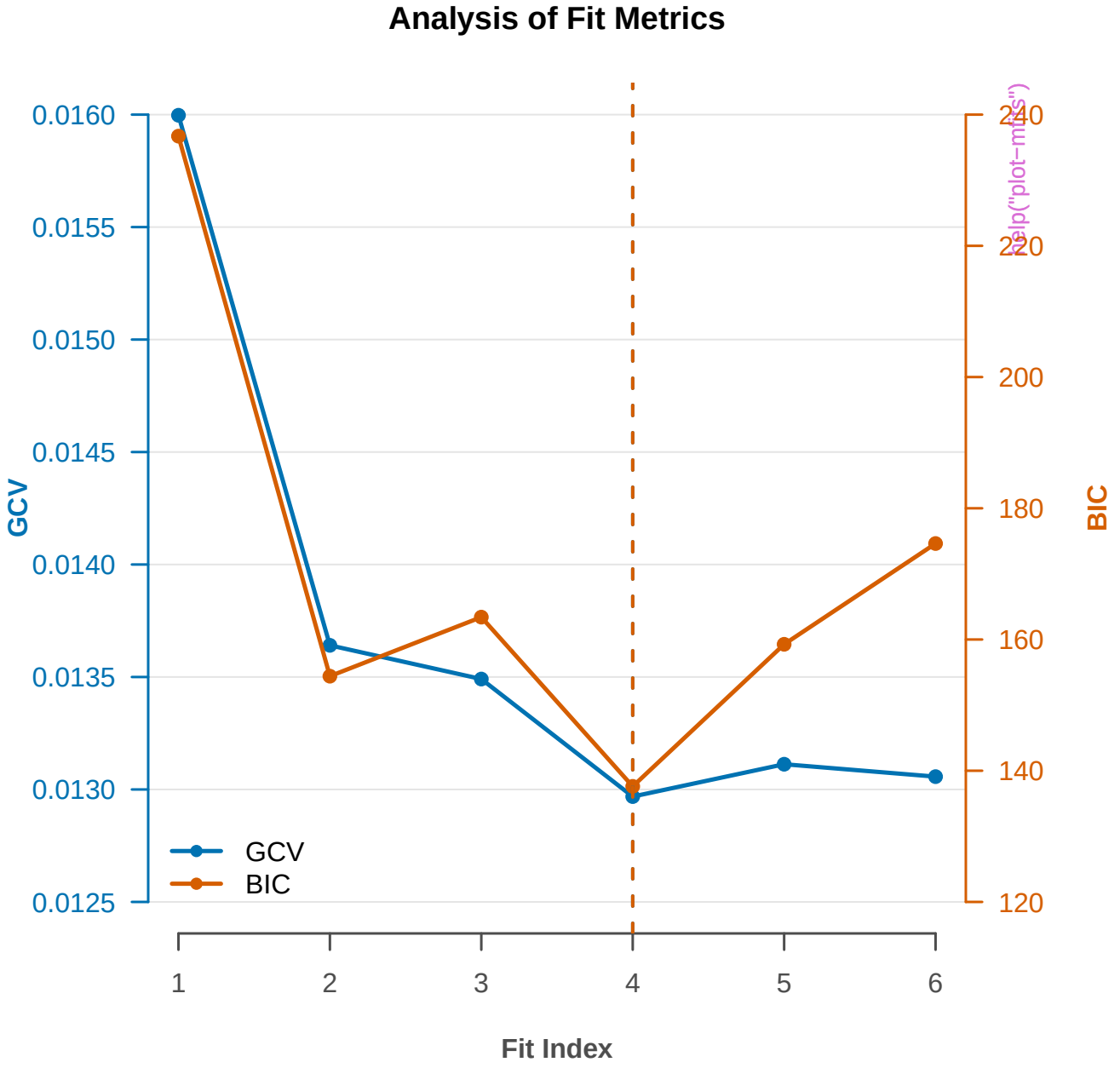
Prediction error

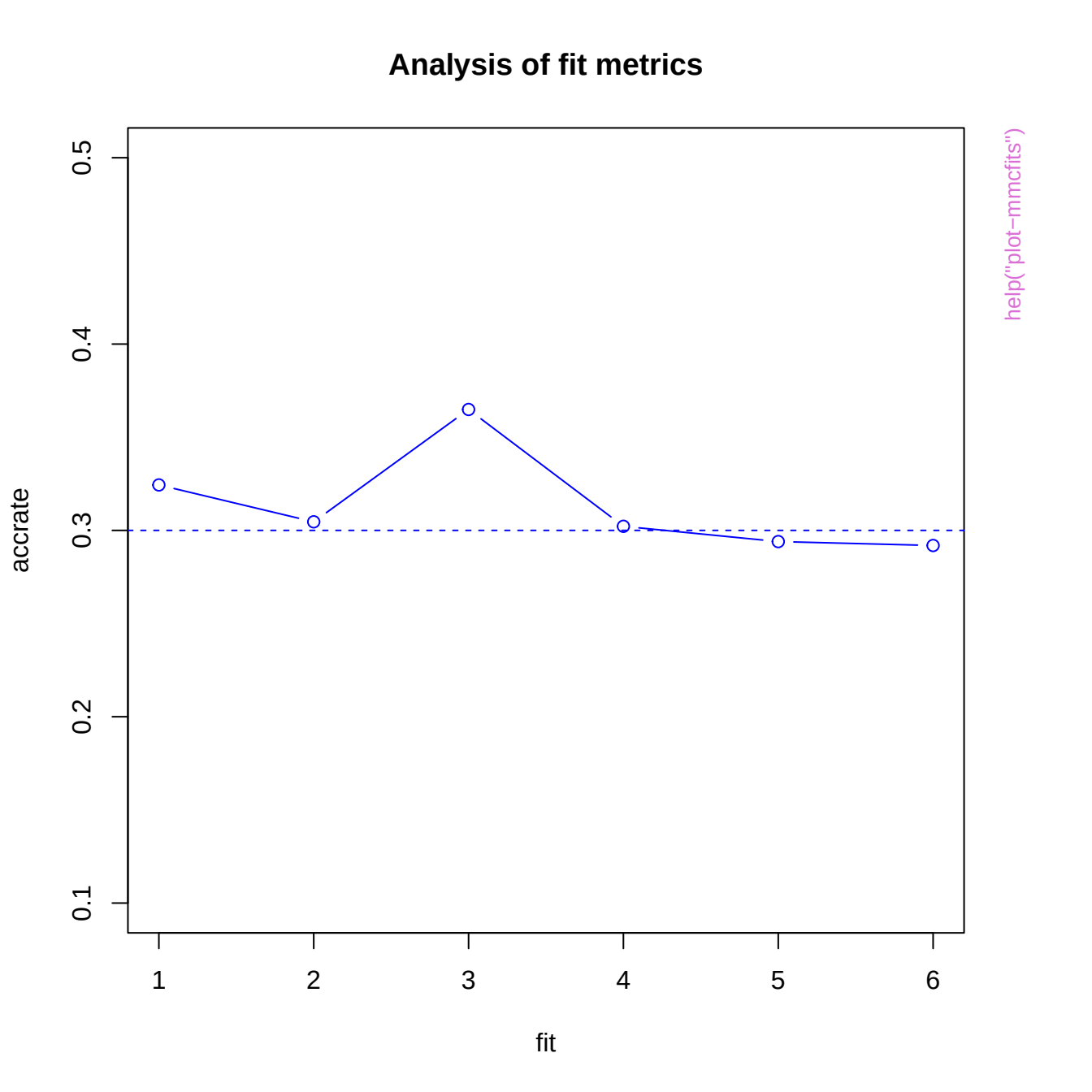


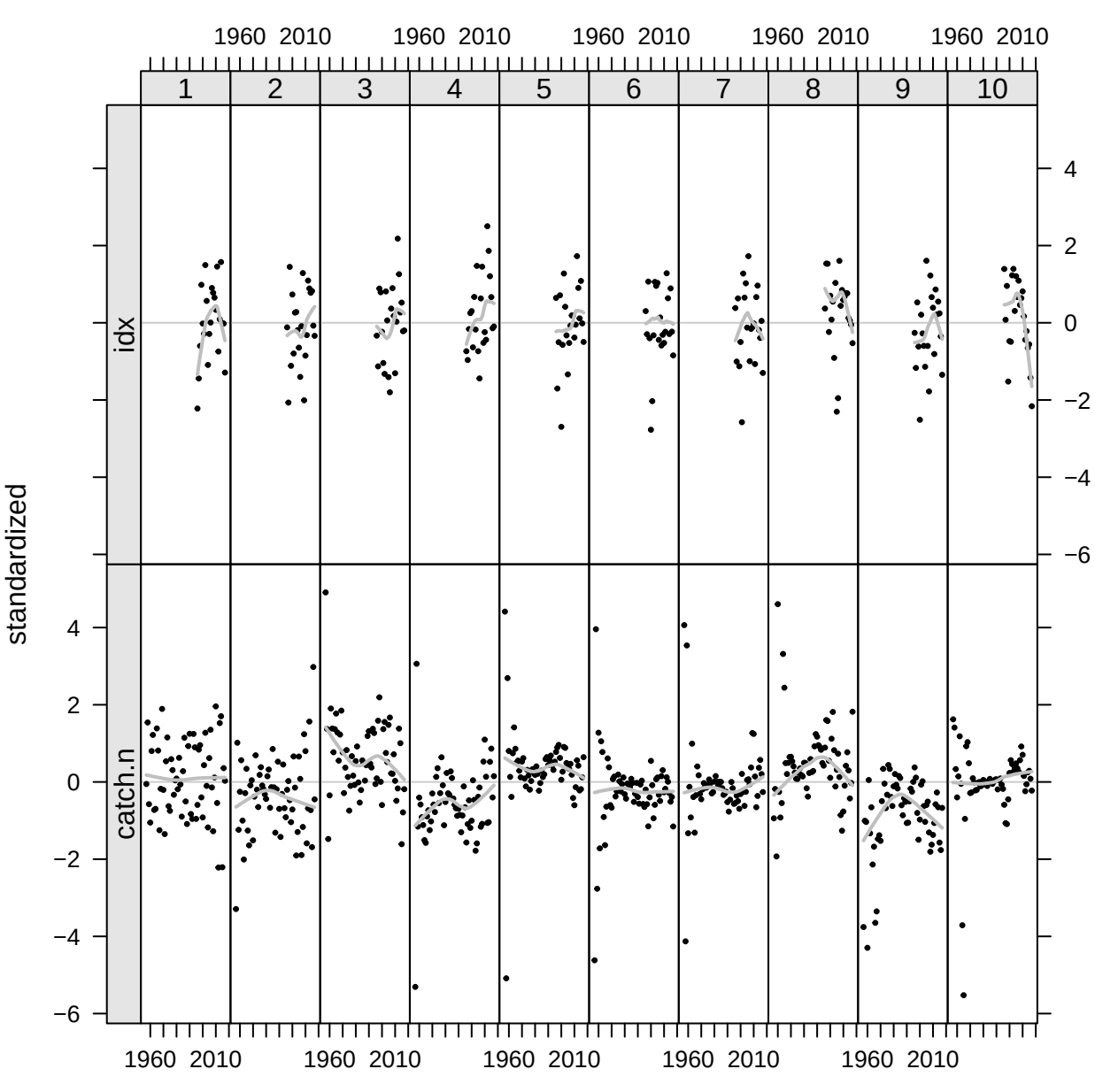
Raw residuals (deviances)

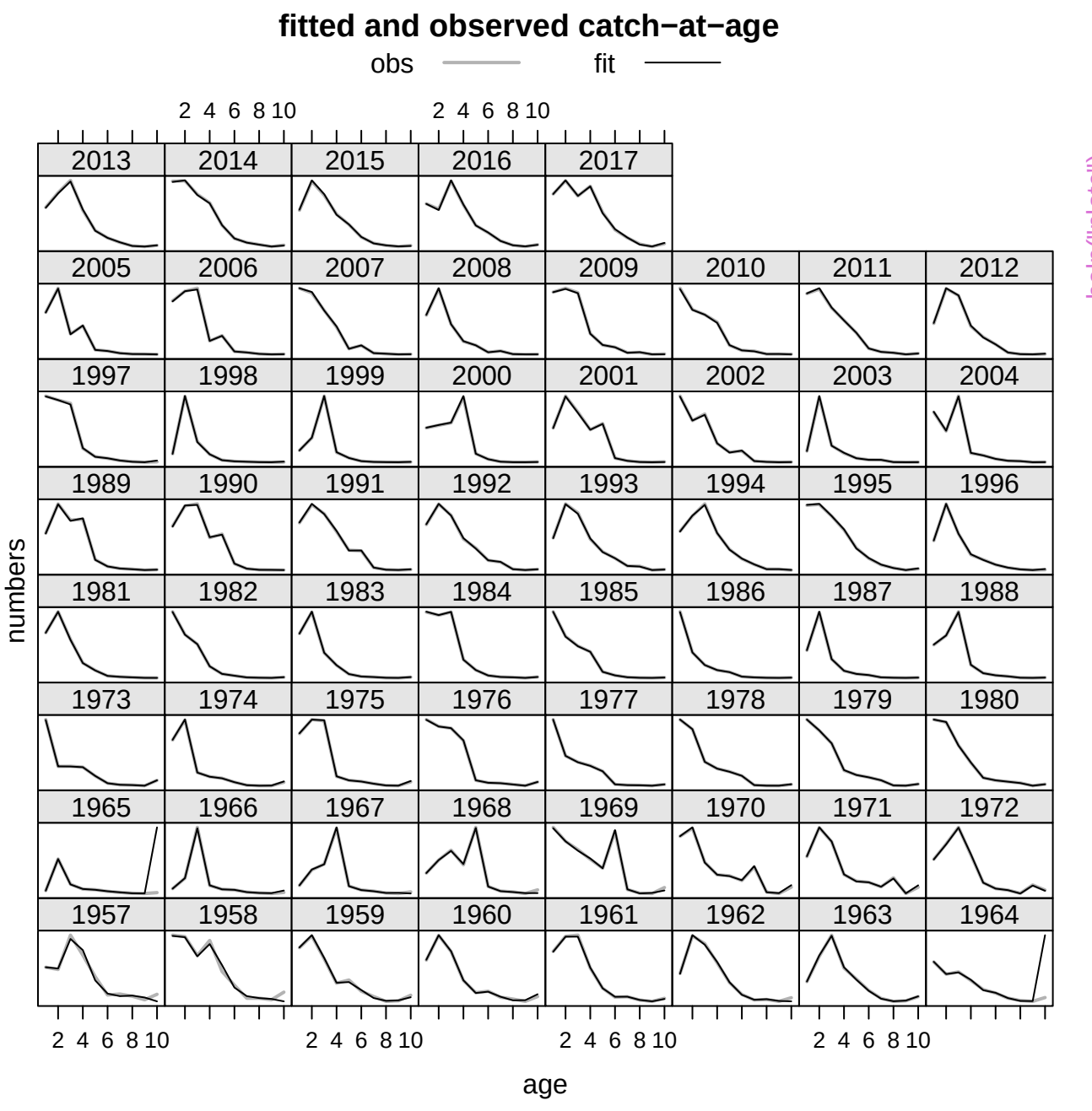


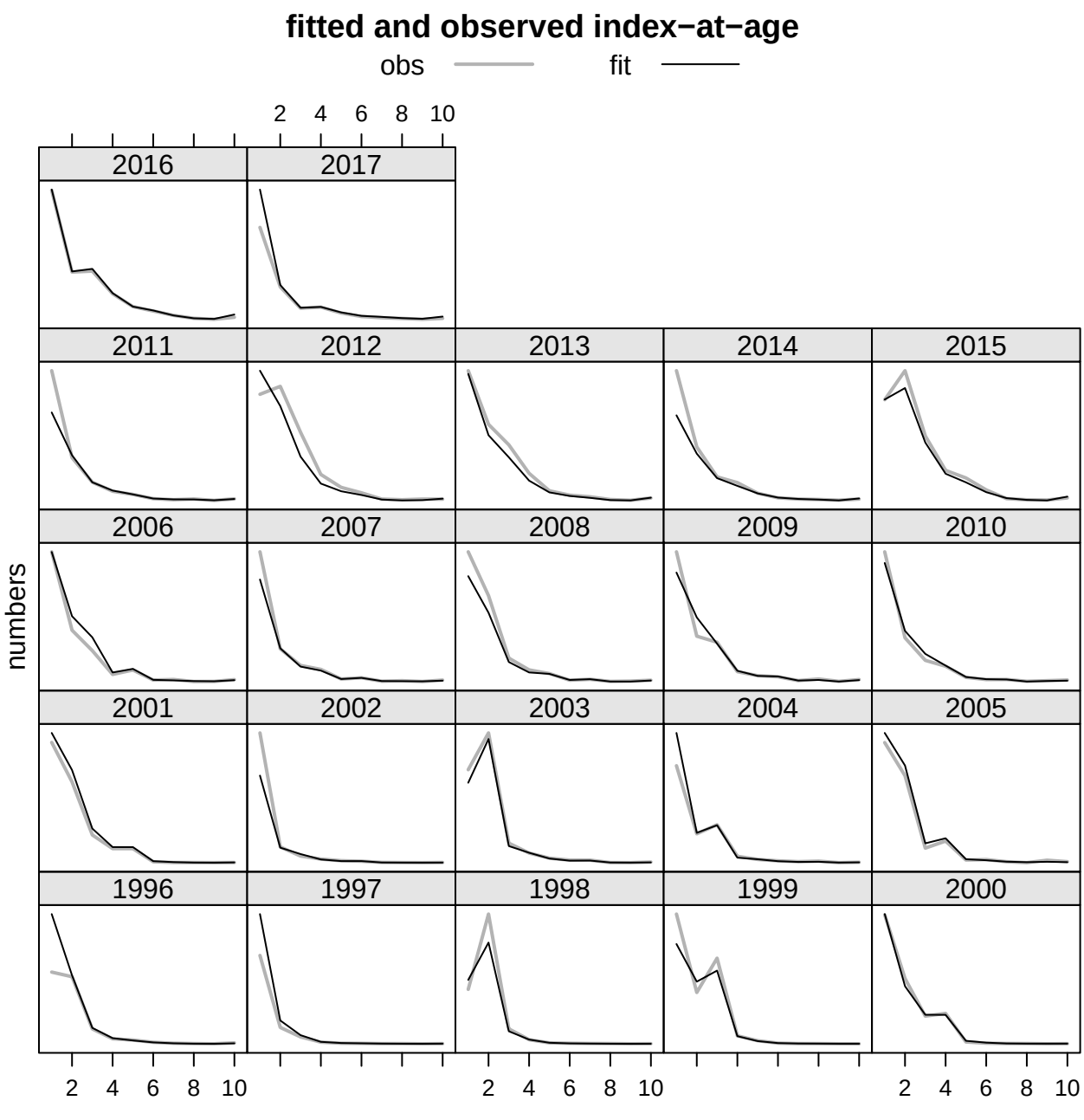
(shaded area = CI 80%, dashed line = median, solid line = observed)


Analysis of Fit Metrics
GCV
0.0160
0.0155
0.0150
0.0145
0.0140
0.0135
0.0130
0.0125
BIC
240
220
200
180
160
140
120
help("plot-mts")
GCV
BIC
1
2
3
4
5
6
Fit Index


Analysis of fit metrics
accrate
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
6
fit
help("plot-mmcfits")

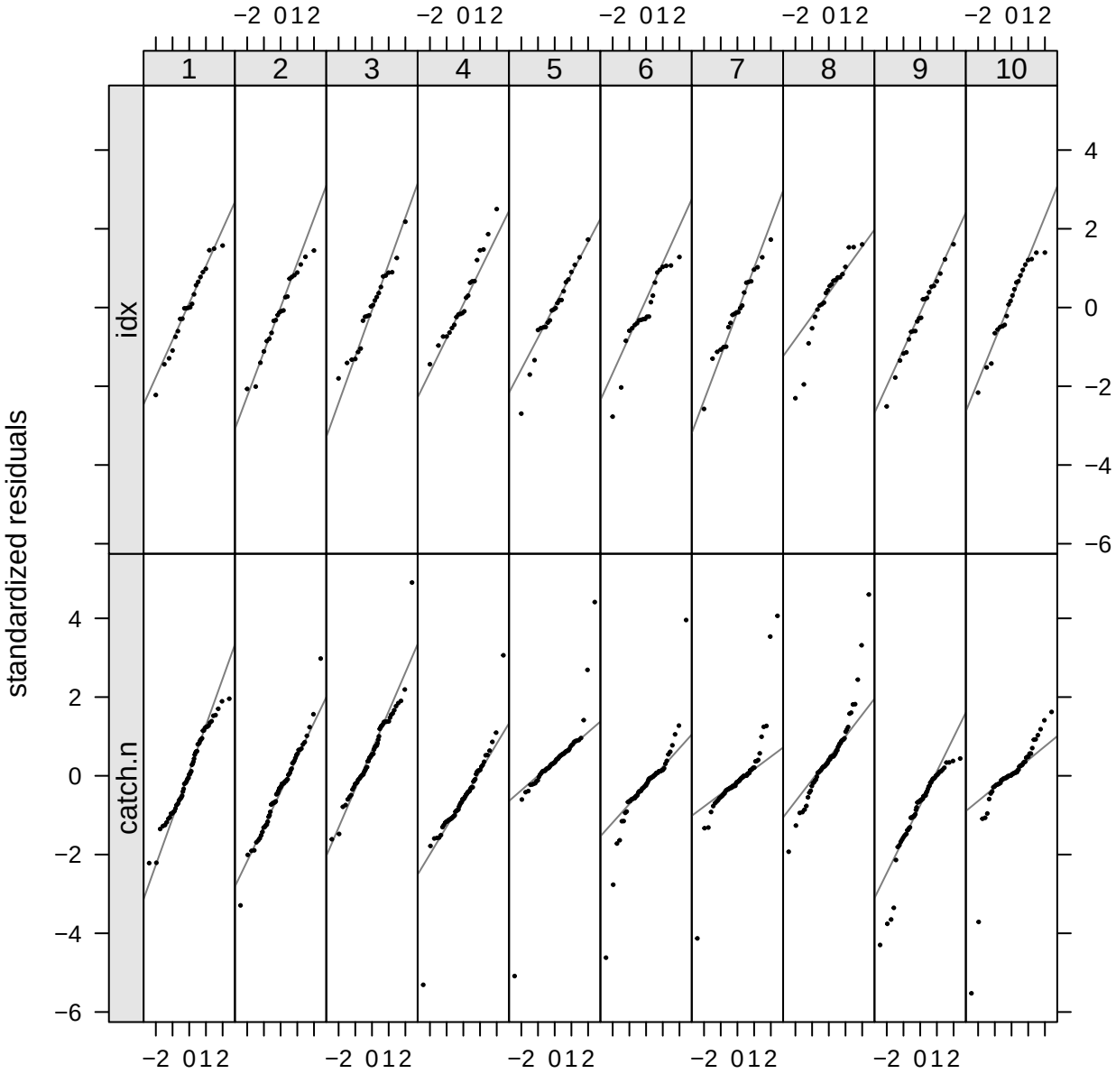

log residuals of catch and abundance indices by year
1960 2010
1 2 3 4 5 6 7 8 9 10
idx
catch.n
standardized
4 2 0 −2 −4 −6
help("plot−res")


fitted and observed catch−at−age
obs
fit
2013
2014
2015
2016
2017
2005
2006
2007
2008
2009
2010
2011
2012
1997
1998
1999
2000
2001
2002
2003
2004
1989
1990
1991
1992
1993
1994
1995
1996
1981
1982
1983
1984
1985
1986
1987
1988
1973
1974
1975
1976
1977
1978
1979
1980
1965
1966
1967
1968
1969
1970
1971
1972
1957
1958
1959
1960
1961
1962
1963
1964
2 4 6 8 10
numbers
age
help("plotc")


fitted and observed index−at−age
obs
fit
numbers
2016
2017
2011
2012
2013
2014
2015
2006
2007
2008
2009
2010
2001
2002
2003
2004
2005
1996
1997
1998
1999
2000
2 4 6 8 10

help("ploti")

quantile−quantile plot of log residuals of catch and abundance indices



help("qqmath−methods")

help("wireframe")

